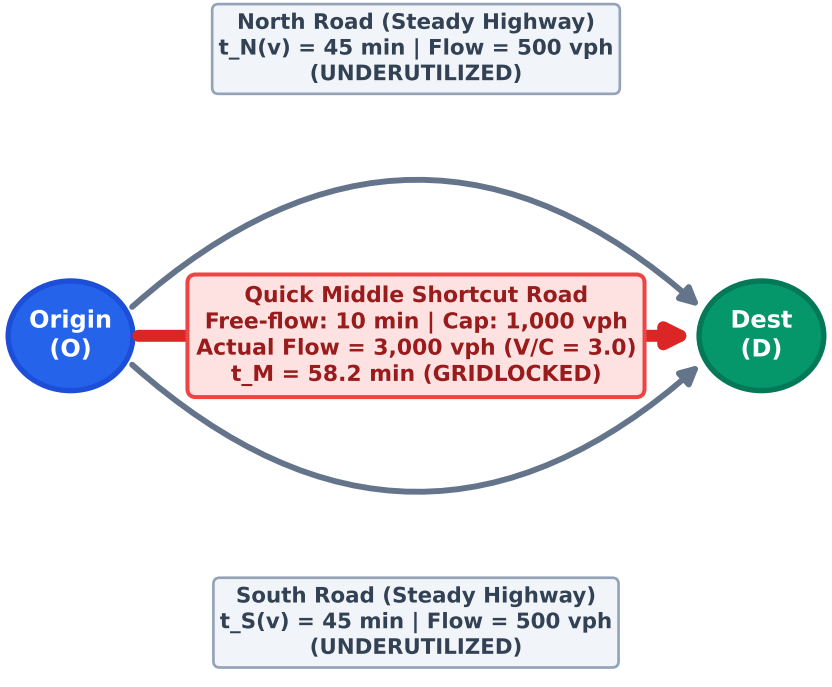
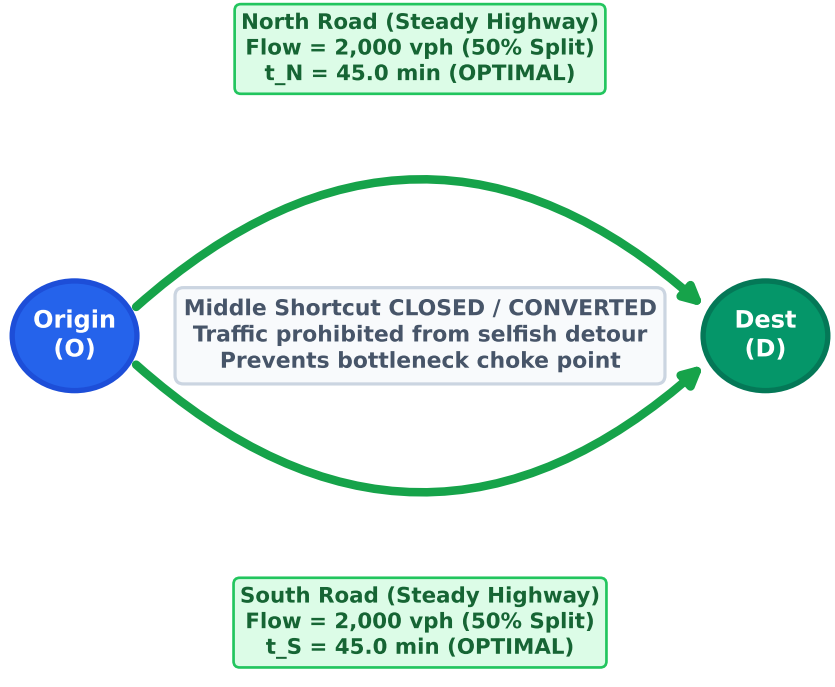
Panel A: User Equilibrium (Middle Quick Road OPEN)
Selfish Routing Overwhelms Shortcut -> Massive Congestion



Result: Total Demand = 4,000 vph | Average Commute = 54.9 min
Selfish choice draws 75% of traffic into the bottleneck!

Panel B: Social Optimum (Middle Road CLOSED / 1-WAY)
Forces 50/50 Split Across Steady Roads -> >20% Faster!



Result: Total Demand = 4,000 vph | Average Commute = 45.0 min
Net Systemic Savings: 9.9 min per vehicle (>20% Travel Time Reduction!)